

MAGDEBURG, Germany

WMO: 103610

Lat: 52.1028N

Lon: 11.5828E

Elev: 279

StdP: 14.55

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.9	14.1	3.5	6.6	12.1	8.8	8.7	15.1	27.7	44.4	24.4	41.0	3.9	60	0.550

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	85.5	66.6	82.1	65.3	78.9	63.8	68.5	81.1	66.7	78.4	65.2	76.1	6.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.3	91.0	72.7	62.3	85.0	71.2	60.5	79.6	70.1	32.8	80.9	31.4	78.3	30.2	75.9	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.6	18.5	16.1	7.5	92.3	10.7	3.3	-0.3	94.7	-6.5	96.7	-12.5	98.6	-20.3	101.0
			6.9	72.4	10.5	1.8	-0.7	73.7	-6.8	74.7	-12.7	75.7	-20.4	77.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.6	33.1	32.5	40.5	46.7	55.8	60.7	64.7	64.4	57.7	49.8	40.4	35.9	(6)
	DBStd	13.43	10.04	9.37	6.88	6.54	6.48	6.19	5.72	5.56	4.96	6.59	6.64	7.18	(7)
	HDD50	2290	525	490	298	140	21	1	0	0	2	84	291	438	(8)
	HDD65	6150	989	909	758	550	289	160	79	81	225	472	737	901	(9)
	CDD50	1786	0	1	5	40	202	321	454	446	233	78	5	1	(10)
	CDD65	172	0	0	0	0	5	30	68	62	6	1	0	0	(11)
	CDH74	2042	0	0	0	2	122	356	795	668	89	10	0	0	(12)
	CDH80	535	0	0	0	0	7	83	235	198	11	1	0	0	(13)
(14) Wind	WSAvg	7.1	8.5	7.3	7.9	7.1	6.2	6.3	6.2	5.9	6.7	6.9	7.8	8.0	(14)
(15) Precipitation	PrecAvg	20.8	1.5	1.1	1.3	1.1	2.3	2.0	2.5	2.3	1.9	1.5	1.6	1.6	(15)
	PrecMax	29.6	2.8	2.1	3.1	3.0	6.3	4.3	4.0	4.6	5.1	4.1	3.5	3.2	(16)
	PrecMin	15.6	0.4	0.3	0.2	0.1	0.5	0.7	0.5	0.7	0.4	0.4	0.0	0.5	(17)
	PrecStd	3.5	0.7	0.6	0.8	0.6	1.4	0.9	1.1	1.0	1.1	0.9	0.8	0.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.2	58.8	64.5	73.0	80.8	87.8	89.1	90.8	81.6	73.6	59.0	54.5	(19)
		MCWB	47.8	51.4	52.3	56.4	62.1	67.4	67.7	68.7	64.7	62.2	51.5	49.2	(20)
	2%	DB	49.3	52.4	57.7	67.5	77.7	82.1	85.7	85.3	76.3	66.8	54.8	51.5	(21)
		MCWB	45.9	47.4	49.5	54.1	60.2	65.4	66.9	66.1	62.6	58.1	50.3	47.4	(22)
	5%	DB	47.1	47.5	53.7	63.3	74.0	78.1	82.4	80.9	71.4	62.7	52.2	49.1	(23)
		MCWB	43.8	44.0	47.0	52.0	59.0	63.2	66.0	64.9	60.6	55.7	48.6	45.9	(24)
	10%	DB	44.8	43.9	50.5	58.9	70.1	74.0	78.7	77.3	67.3	59.8	50.0	46.6	(25)
		MCWB	41.6	40.4	45.3	49.4	57.5	61.6	64.3	63.3	58.6	54.2	47.0	43.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.6	51.9	53.1	57.9	64.7	70.4	71.5	70.5	66.5	63.0	52.8	49.9	(27)
		MCDB	51.5	57.7	61.0	70.3	76.0	82.1	85.6	87.9	78.7	71.6	56.6	53.7	(28)
	2%	WB	46.2	47.4	50.2	55.2	62.2	67.2	68.8	67.8	63.8	59.5	50.7	47.9	(29)
		MCDB	49.5	51.4	57.3	65.5	74.6	78.7	81.0	80.1	73.3	65.9	54.1	51.2	(30)
	5%	WB	44.0	44.0	47.9	52.7	60.0	65.0	66.9	66.2	61.7	56.7	49.1	46.1	(31)
		MCDB	46.8	47.4	52.7	61.3	71.7	75.1	78.9	77.6	69.4	61.5	52.1	48.9	(32)
	10%	WB	41.7	40.6	45.7	50.5	57.9	62.6	65.0	64.6	59.9	54.7	47.0	43.9	(33)
		MCDB	44.5	43.6	50.2	58.2	69.0	72.0	76.2	75.1	66.6	59.0	49.9	46.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	11.7	13.4	17.1	19.5	18.2	19.8	20.0	17.0	14.9	10.4	8.2	(35)
		MCDBR	10.0	15.6	19.3	24.1	26.6	26.0	27.8	27.3	24.1	20.0	13.1	10.6	(36)
	5% WB	MCWBR	8.4	11.6	12.7	12.9	13.2	12.2	12.4	12.2	13.0	12.4	10.4	8.5	(37)
		MCWBR	10.3	15.6	18.5	21.5	24.2	23.0	24.6	22.9	20.7	18.6	12.6	10.0	(38)
(40) Clear-Sky Solar Irradiance	taub		0.315	0.337	0.375	0.411	0.413	0.409	0.428	0.420	0.384	0.358	0.335	0.309	(40)
		taud	2.375	2.355	2.260	2.207	2.239	2.271	2.229	2.265	2.355	2.410	2.406	2.372	(41)
	Ebn at Noon		217	246	256	261	266	268	260	255	252	234	207	198	(42)
		Edh at Noon	19	26	34	41	41	40	42	38	31	24	19	16	(43)
(44) All-Sky Solar Radiation	RadAvg		216	437	777	1267	1556	1678	1575	1341	979	558	254	165	(44)
	RadStd		23	62	83	152	180	135	214	117	117	60	30	16	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.72	N/A	N/A	N/A	N/A

Nomenclature: See separate page